

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

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MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

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COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

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MINISTRY OF HIGHER EDUCATION

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NATIONAL COMMISSION FOR THE ORGANISATION OF
HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma 2023 Session

Speciality/Option: SOFTWARE ENGINEERING (SWE)

Paper : Information Systems

Duration : 3 hours

Credit : 6

ANSWER ALL QUESTIONS

SECTION A: SYSTEM ARCHITECTURE (40 marks)

- 1) What is an information system in an organization? **4mks**
- 2) Name the five important components of an information system? **4mks**
- 3) Draw an diagram to show relation of information system to the levels of organization. **4mks**
- 4) What are the functions of an information system? **4mks**
- 5) Name the steps involved in the development of an information system. **4mks**
- 6) Define the client server architecture concept. **4mks**
- 7) What are the 3 functions of client-server? **4mks**
- 8) What are the two broad classes of middleware in client server environment? **4mks**
- 9) What are the five major technologies that can be used to create Client/Server applications in client server environment? **4mks**
- 10) What is public cloud and private cloud? **4mks**

SECTION B: OPERATING SYSTEMS (40 marks)

I. MCQs (20 Marks)

Each Correct Answer Carries One (01)

1. CPU fetches the instruction from memory according to the value of:
 - A. program counter
 - B. status register
 - C. instruction register
 - D. program status word

2. A memory buffer used to accommodate a speed differential is called:
- A. stack pointer
 - B. cache
 - C. accumulator
 - D. disk buffer
3. Which one of the following is the address generated by CPU?
- A. physical address
 - B. absolute address
 - C. logical address
 - D. none of the mentioned
4. Run time mapping from virtual to physical address is done by:
- A. memory management unit
 - B. CPU
 - C. PCI
 - D. none of the mentioned
5. Memory management technique in which system stores and retrieves data from secondary storage for use in main memory is called:
- A. fragmentation
 - B. paging
 - C. mapping
 - D. none of the mentioned
6. Effective access time is directly proportional to:
- A. page-fault rate
 - B. hit ratio
 - C. memory access time
 - D. none of the mentioned
7. In FIFO page replacement algorithm, when a page must be replaced:
- A. oldest page is chosen
 - B. newest page is chosen
 - C. random page is chosen
 - D. none of the mentioned

8. Which algorithm chooses the page that has not been used for the longest period of time whenever the page required to be replaced?
- A. first in first out algorithm
 - B. additional reference bit algorithm
 - C. least recently used algorithm
 - D. counting based page replacement algorithm
9. A process is thrashing if:
- A. it is spending more time paging than executing
 - B. it is spending less time paging than executing
 - C. page fault occurs
 - D. swapping can not take place
10. Working set model for page replacement is based on the assumption of:
- A. modularity
 - B. locality
 - C. globalization
 - D. random access
11. The operating system is :
- A. in the low memory
 - B. in the high memory
 - C. either a or b (depending on the location of interrupt vector)
 - A. D. None of these
12. In contiguous memory allocation :
- A. each process is contained in a single contiguous section of memory
 - B. all processes are contained in a single contiguous section of memory
 - C. the memory space is contiguous
 - D. None of these
13. The relocation register helps in :
- A. providing more address space to processes
 - B. a different address space to processes
 - C. to protect the address spaces of processes
 - A. D. None of these

14. With relocation and limit registers, each logical address must be _____ the limit register.
- A. less than
 - B. equal to
 - C. greater than
 - D. None of these
15. The operating system and the other processes are protected from being modified by an already running process because :
- A. they are in different memory spaces
 - B. they are in different logical addresses
 - C. they have a protection algorithm
 - D. every address generated by the CPU is being checked against the relocation and limit registers
16. Transient operating system code is code that :
- A. is not easily accessible
 - B. comes and goes as needed
 - C. stays in the memory always
 - D. never enters the memory space
17. Using transient code, _____ the size of the operating system during program execution.
- A. increases
 - B. decreases
 - C. changes
 - D. maintains
18. When memory is divided into several fixed sized partitions, each partition may contain ____.
- A. exactly one process
 - B. at least one process
 - C. multiple processes at once
 - D. None of these
19. In fixed sized partition, the degree of multiprogramming is bounded by _____.
- A. the number of partitions
 - B. the CPU utilization
 - C. the memory size
 - D. All of these

20. The first fit, best fit and worst fit are strategies to select a _____.

- A. process from a queue to put in memory
- B. processor to run the next process
- C. free hole from a set of available holes
- D. All of these

II. Structural Questions (20 Marks)

- 1. What does a systems analyst do? **4 marks**
- 2. Name the roles of a systems analyst. **4 marks**
- 3. What are the characteristics of a good system analyst? **4 marks**
- 4. What are the four skills of a systems analyst? **4 marks**
- 5. Name the four types of feasibility study. **4 marks**

SECTION C: PROJECT MANAGEMENT AND LEGAL REGULATIONS (20 marks)

- 1. Define the following terms:
 - a) Project **1 mark**
 - b) Software project manager **1 mark**
- 2. Software project management comprises of a number of activities, which contains planning of project, deciding scope of software product, estimation of cost in various terms, scheduling of tasks and events, and resource management.
 - a) What are the activities that are carried out during project scope management activities? **4 marks**
 - b) What is usually estimated during project estimation phase? **7 marks**
 - c) Discuss two types of Project Estimation Techniques. **2 marks**
- 3.
 - a) What is Project Scheduling? **2 marks**
 - b) What are the activities that are carried out during Project Scheduling? **4 marks**